Credit Risk Analysis Dashboard *17,650 Applicants | Default Rate: 7.03%*

High Risk | Low Risk | Medium Risk

17,650
Total Applicants

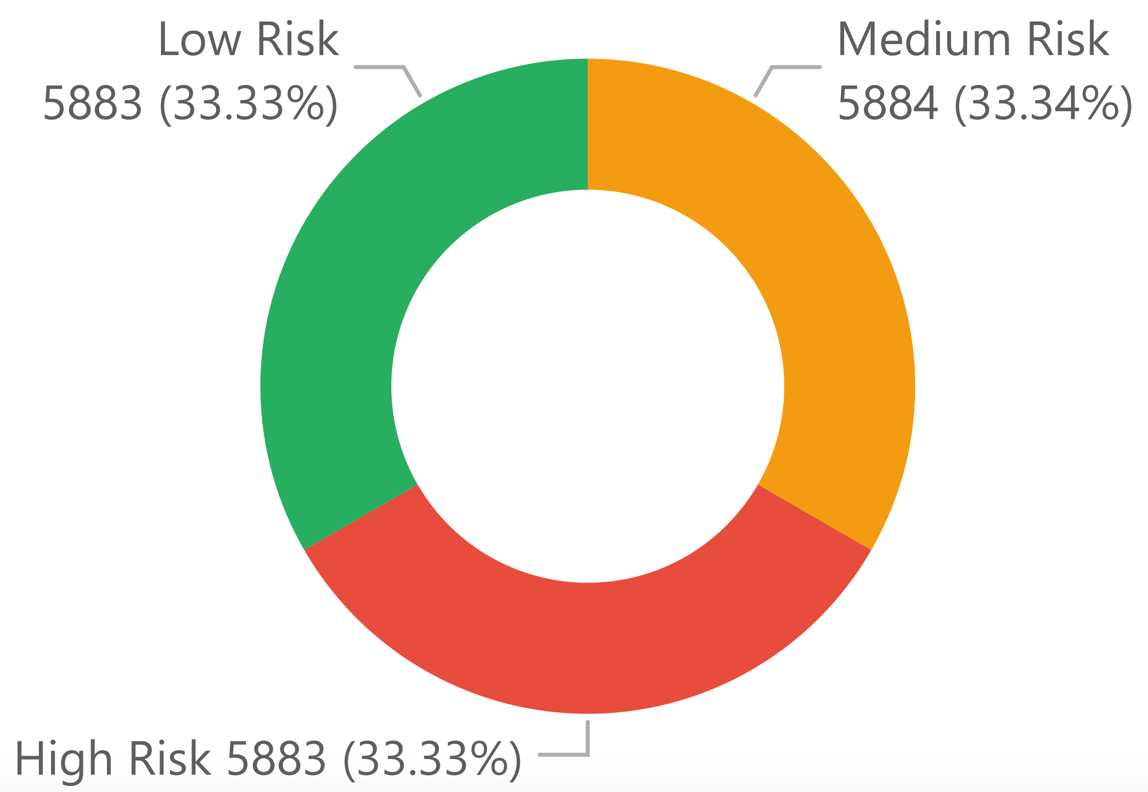
7.03%
Default rate %

6K
High Risk Count

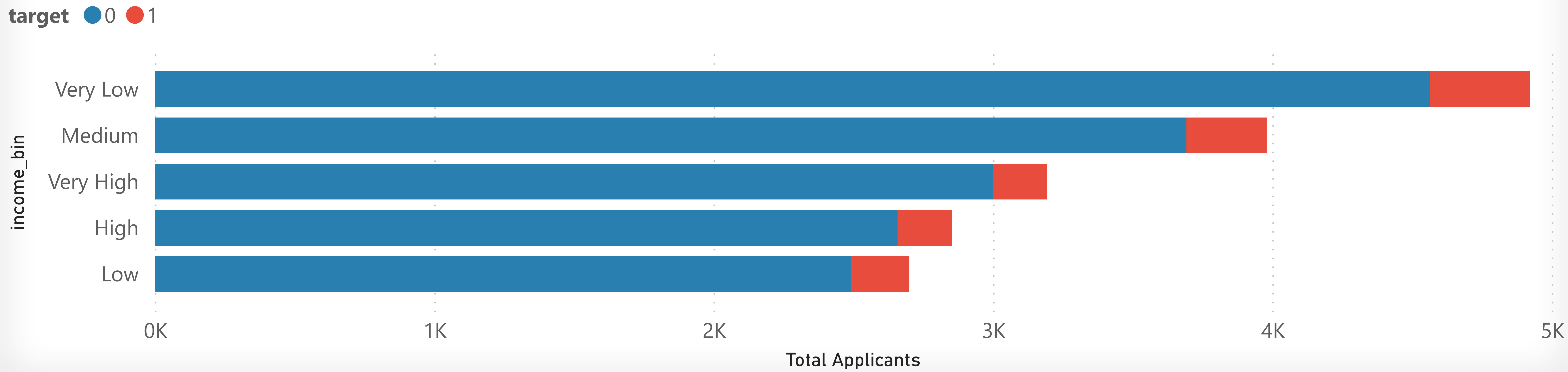
0.45
Avg Risk Score

33.33%
Portfolio at Risk %

Portfolio Risk Distribution



Default Rate By Income Band



Risk Score Distribution

Risk Segmentation Analysis

Comparing applicant profiles across Low, Medium, and High risk groups

High Risk | Low Risk | Medium Risk

17,650
Total Applicants

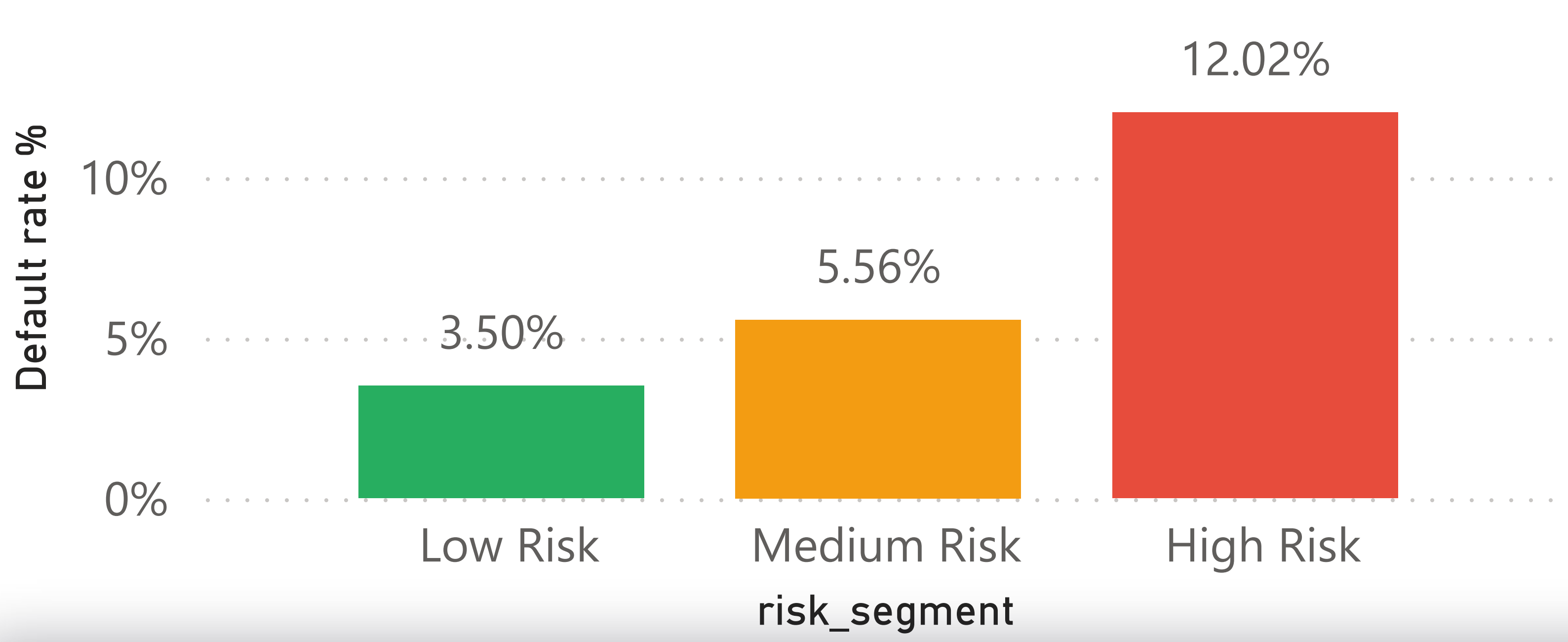
7.03%
Default rate %

5.883K
High Risk Count

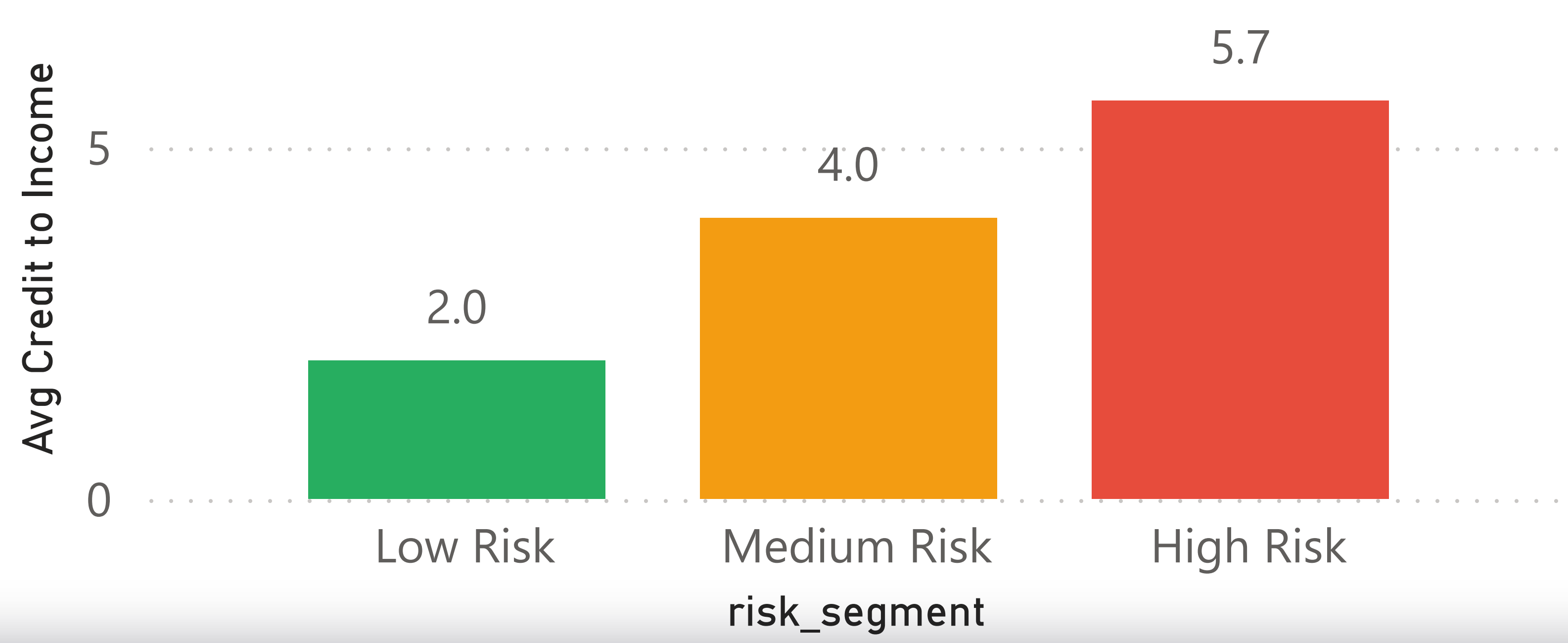
0.45
Avg Risk Score

33%
Portfolio at Risk %

Default Rate by Risk Segment



Average Credit-to-Income Ratio



risk_segment	Applicants Count	Defaults Count	Default rate %
High Risk	5883	707	12.02%
Medium Risk	5884	327	5.56%
Low Risk	5883	206	3.50%

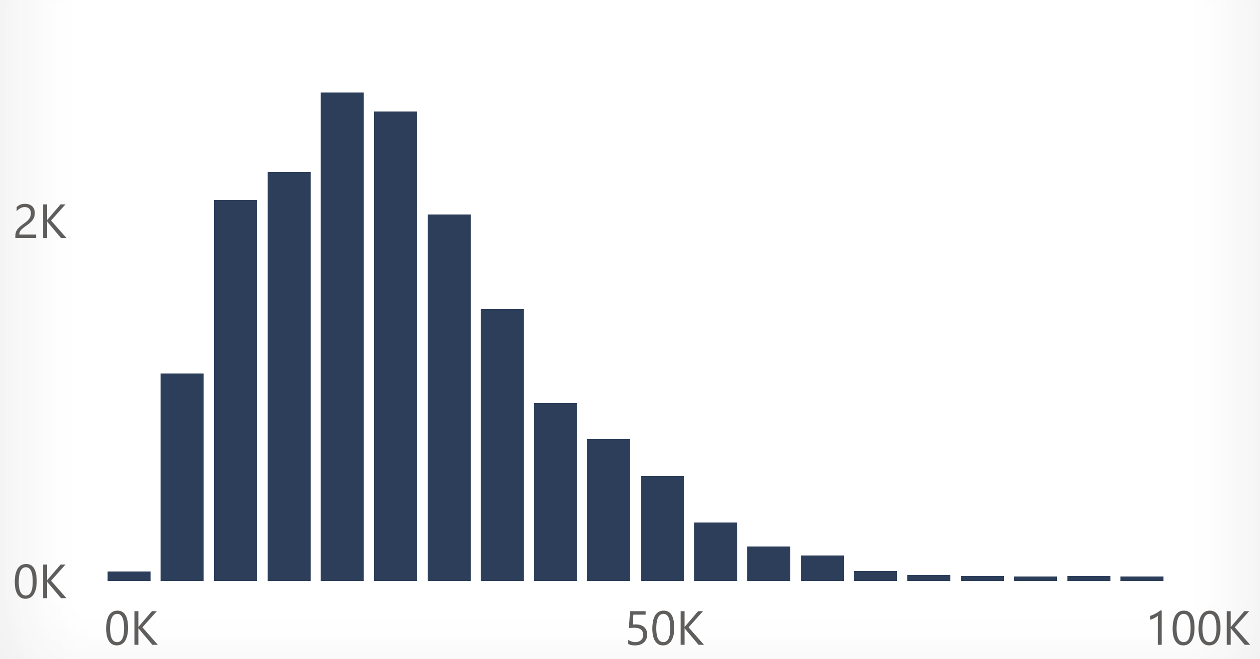
Credit vs Credit-to-Income by Risk Segment

*High-risk applicants default at ~3.4× the rate of low-risk applicants, driven by significantly higher credit-to-income ratios.

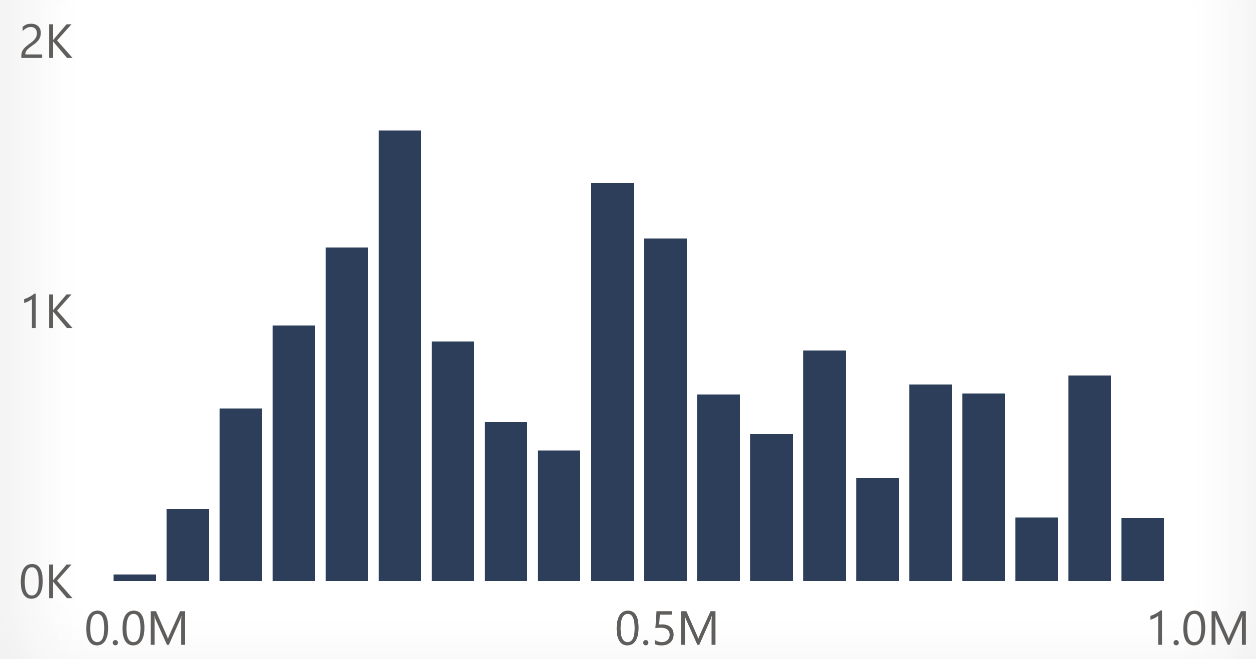
EDA — Exploratory Data Analysis

Distribution patterns and feature correlations across 17,650 applicants

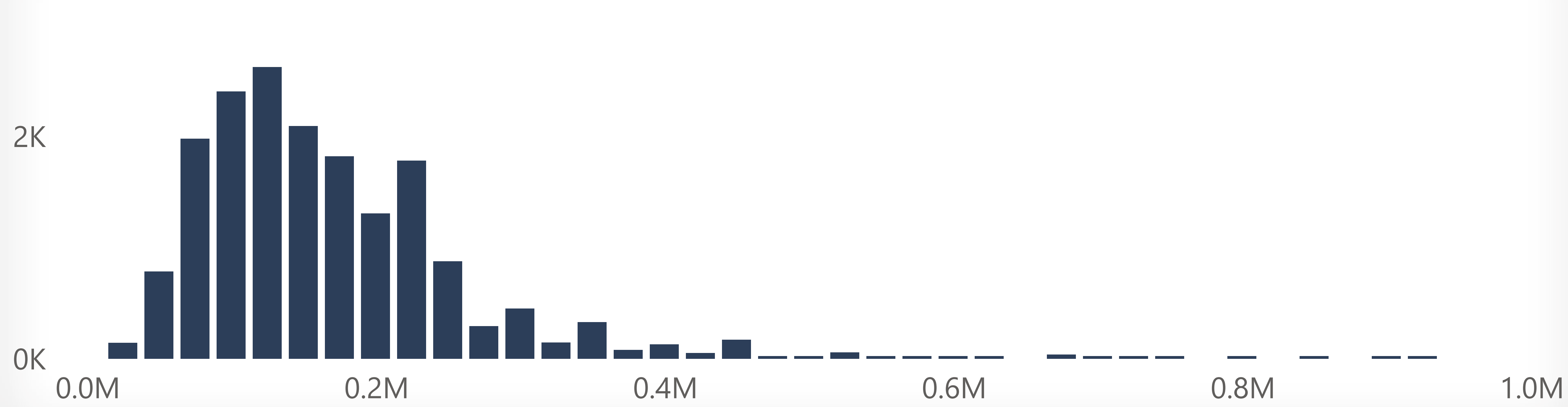
Annuity Distribution



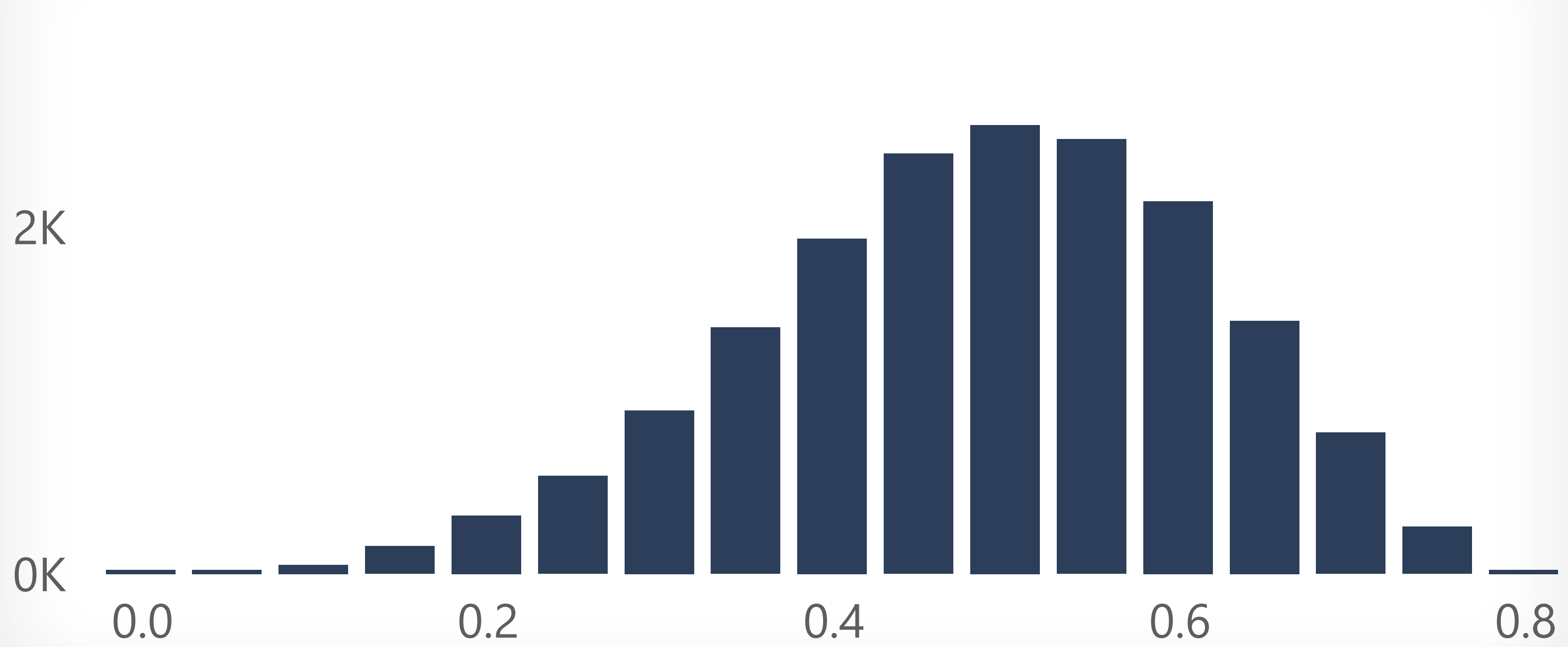
Credit Amount Distribution



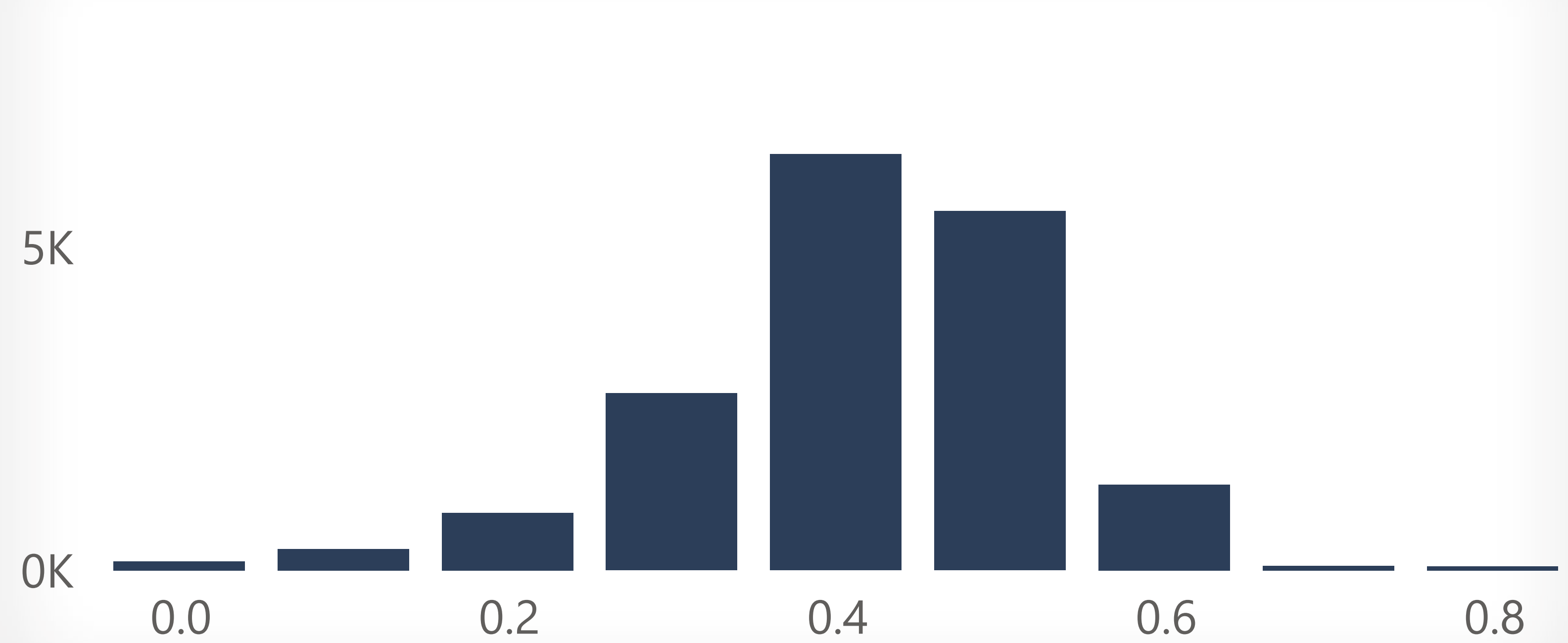
Income Distribution



External Score Distribution

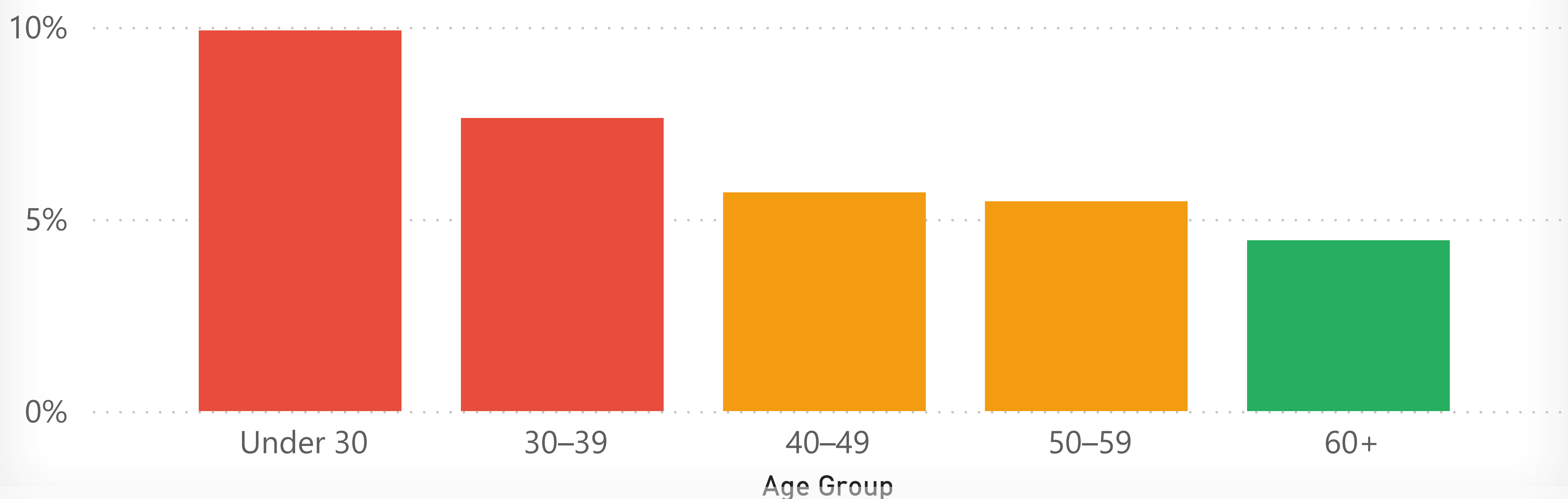


Risk Score Distribution

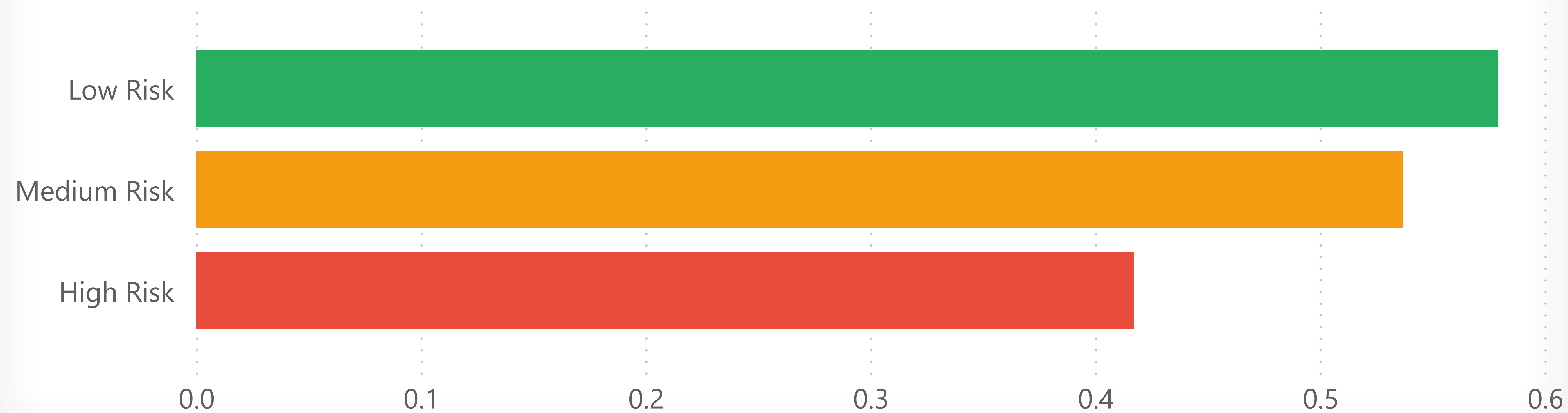


Metric	Credit	CTI	External Score	Income	Risk Score
Credit	1.00			0.04	0.11
CTI		1.00			0.16
External Score			1.00		-0.10
Income	0.04			1.00	
Risk Score	0.11	0.16	-0.10		1.00

Default rate % by Age Group



Average Ext Score by Risk Segment

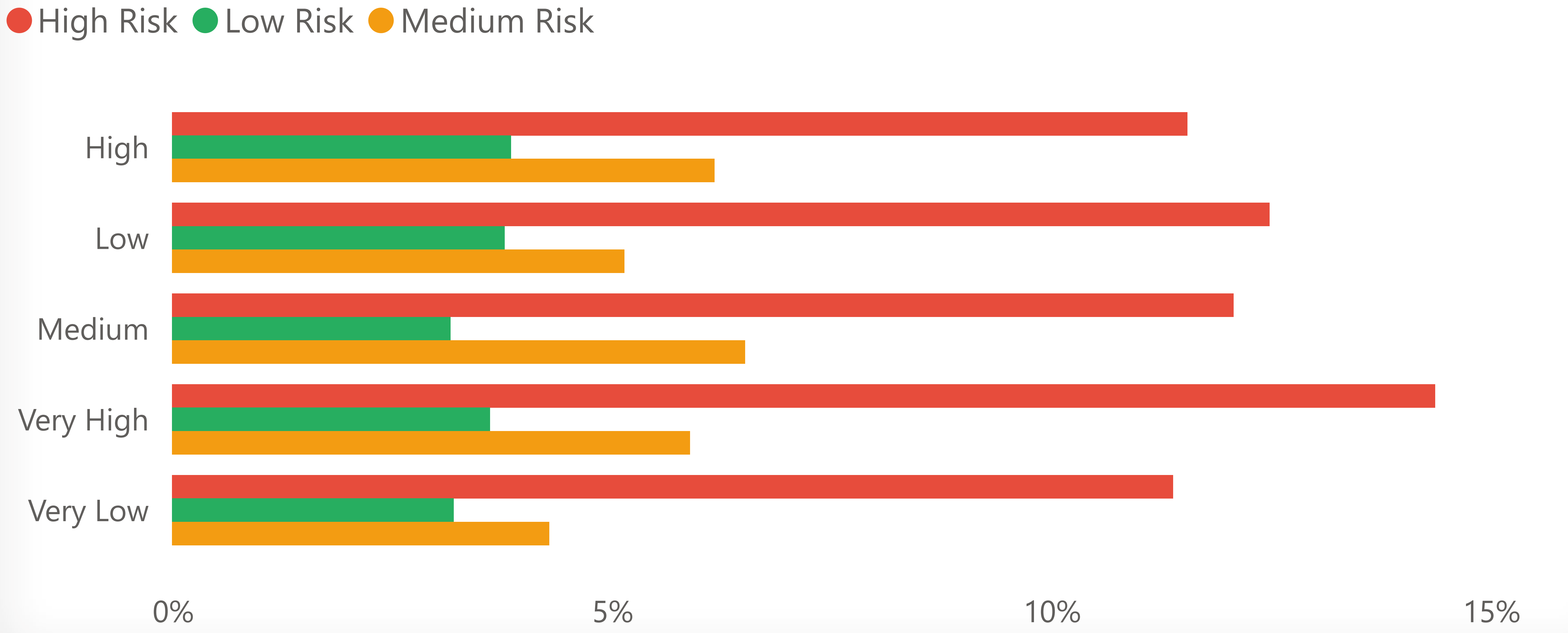


**Younger applicants (Under 30) default at 9.91% — nearly 2.5× the rate of those 60+. High-risk applicants score 25% lower on external credit assessments, confirming EXT_SOURCE as the strongest default predictor in this portfolio.*

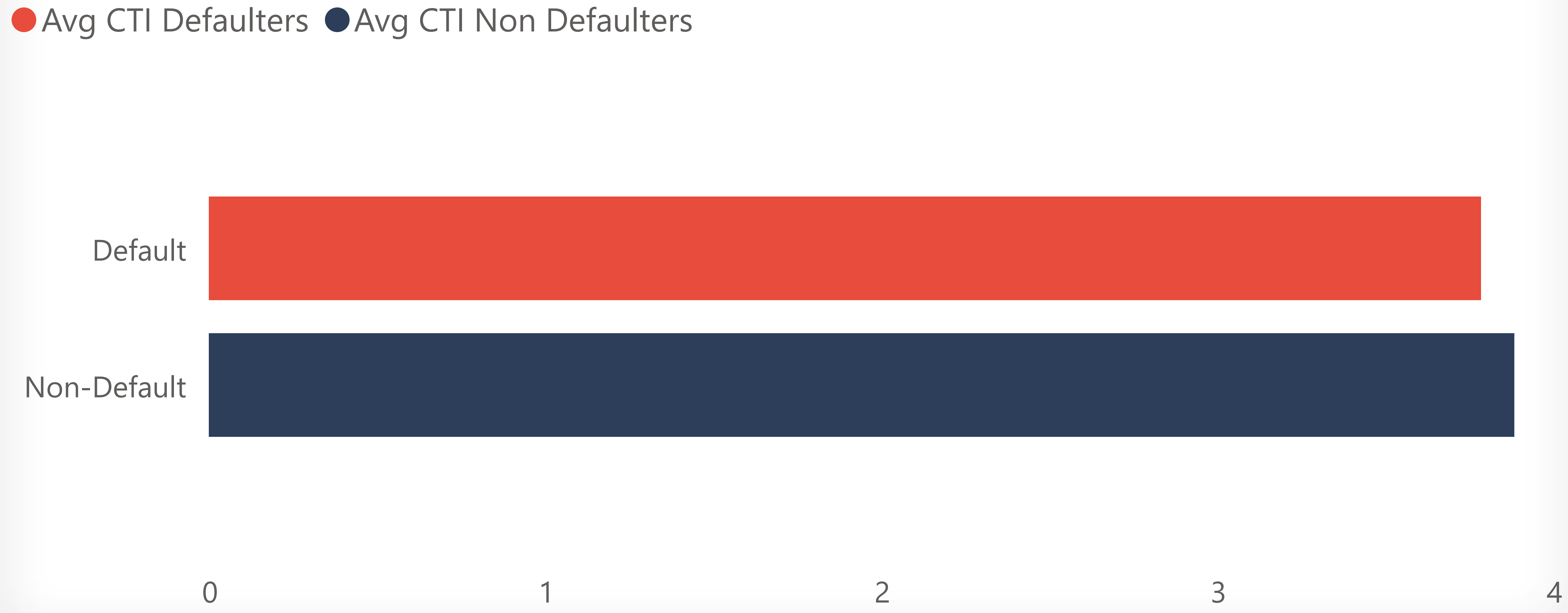
Business Insights

Credit-to-income analysis and risk-income interaction effects

Default Rate by Income and Risk Segment



Avg CTI Defaulters and Avg CTI Non Defaulters by Target Label



risk_segment	25000	50000	75000	100000	125000	150000	175000	200000	2
High Risk	6.58%	10.29%	10.98%	12.88%	12.44%	12.99%	10.77%	11.36%	
Low Risk	4.17%	2.92%	2.73%	4.09%	3.25%	2.85%	3.70%	4.72%	
Medium Risk	2.27%	2.98%	3.45%	5.87%	4.98%	7.12%	5.30%	7.42%	
Total	**4.86%**	**6.50%**	**6.83%**	**8.36%**	**7.40%**	**7.84%**	**6.37%**	**7.74%**	

3.78

Avg CTI Defaulters

3.88

Avg CTI Non Defaulters

-0.10

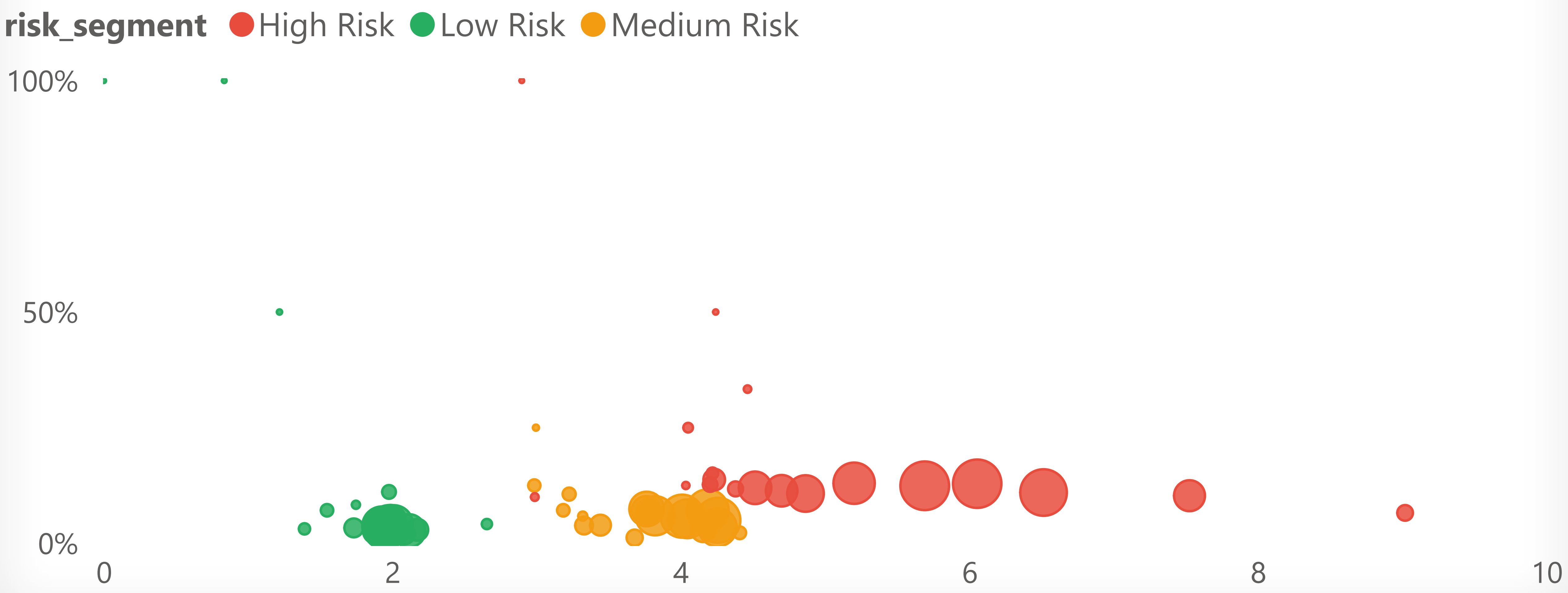
CTI Gap

**Defaulters and non-defaulters show near-identical CTI ratios (3.78 vs 3.88), suggesting CTI alone is insufficient — risk segmentation combined with external scores provides stronger predictive power.*

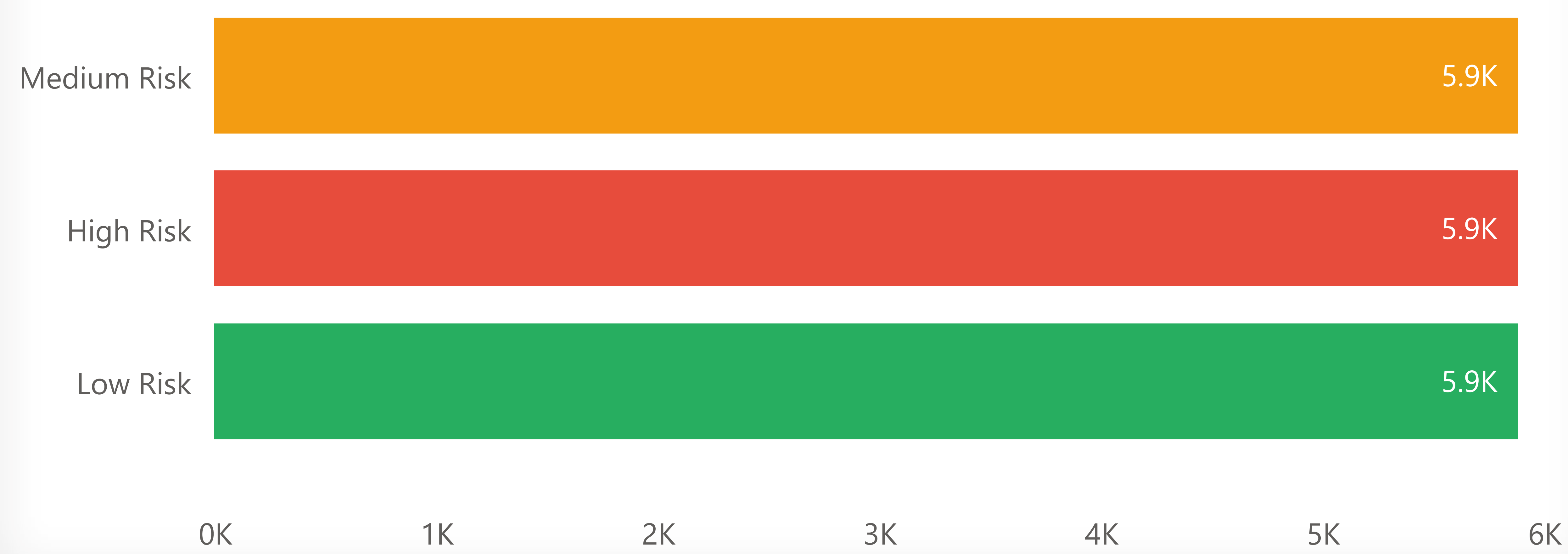
Portfolio Strategy

Risk-based lending recommendations for the Home Credit portfolio

Risk Matrix- Defaukt Rate vs Credit Burden



Portfolio Composition by Risk Segment



Segment Summary Table

risk_segment	Default rate %	Avg Income by Segment	Avg Credit by Segment	Total Applicants
High Risk	12.02%	148,219.39	784,883.69	5883
Low Risk	3.50%	229,846.32	401,167.69	5883
Medium Risk	5.56%	176,033.87	682,926.12	5884
Total	**7.03%**	**184,699.37**	**622,995.89**	**17650**

Portfolio Recommendations

🔴 High Risk (33%) — Restrict credit limits, require collateral, increase monitoring frequency

🟡 Medium Risk (33%) — Conditional approval with income verification and reduced loan tenure

🟢 Low Risk (33%) — Fast-track approvals, offer competitive rates, upsell opportunity

**A risk-tiered lending strategy targeting the High Risk segment for restriction and Low Risk for growth could reduce portfolio default rate from 7.03% to an estimated 4–5% while maintaining approval volumes.*